The Inference Bottleneck: Structural Limits on Observation-to-Action Throughput

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Abstract

We prove that any system with (1) finite-dimensional state, (2) positive observation model, and (3) causal serial interaction has observation-to-action information throughput rate bounded by $(k+1) \cdot R_{\max}$, where k is the treewidth of the system's effective conditional-independence graph and $R_{\max}$ is the maximum per-coordinate information rate. The bound is continuous in time, parametric in k, structural (independent of the observation process), and unconditional: it is a consequence of the data processing inequality, not a complexity-theoretic conjecture. We show that the environment's coupling structure imposes a lower bound on the achievable treewidth of any system that adequately represents it, and define the *effective throughput* — the information about the true environment state reaching actions — which is reduced below the ceiling by a quantifiable model mismatch tax. We further establish that the axioms themselves entail exponential family structure: exactly, via finite sufficiency (Pitman–Koopman–Darmois) for systems performing exact inference; and asymptotically, via the Bernstein–von Mises theorem for systems performing approximate inference, under a physical non-degeneracy condition and a mild local-prior-support assumption. Tractable evaluation of the resulting log-partition function on a graph G necessitates bounded treewidth: $\mathrm{tw}(G) = O(1)$, for both exact evaluation (conditional on FPT $\neq$ #W[1]) and approximate evaluation (conditional on ETH). Thus the exponential-family setting enters as a consequence of the axioms, not as a separate restriction on the model class. For singular models — where the Fisher information is degenerate and the Bernstein–von Mises theorem fails — we establish the same bounded-treewidth conclusion via a backward reduction: any polynomial-time learner achieving the Watanabe-optimal generalization rate for a graphical model with latent variables must implicitly compute the partition function (Watanabe, 2009), and the same complexity-theoretic obstruction applies. For fully observed singular models, we establish the same conclusion via a second backward reduction: tractable comparative Bayesian revision together with a tractable partition anchor reveals the free-energy differences needed to recover the partition function. The discrete per-step bound of the companion paper is recovered as an immediate corollary.

1 Introduction

Motivation. What is the maximum rate at which a tractable reasoning system can convert observations into actions? This paper gives a precise answer. Any system that (1) has finite-dimensional state, (2) maintains a positive observation model, and (3) interacts serially with its environment has instantaneous observation-to-action throughput bounded by $(k+1) \cdot R_{\max}$, where k is the treewidth of the system's effective conditional-independence graph and $R_{\max}$ is the maximum per-coordinate information rate. Every system with coordinates routes information through *some* graph (in the worst case, the complete graph K_N, with treewidth $N-1$). The bound is stated in terms of that graph, and the necessity results (§9) show that tractable inference forces its treewidth to be bounded. The bound is not a performance limitation that better engineering could remove — it is a structural ceiling imposed by the data processing inequality. A system with higher treewidth can push through more information per second, but the computational cost of maintaining that treewidth grows exponentially: each additional unit of k costs an exponential factor in inference time. The result is substrate-independent. It applies to any system satisfying the axioms, whether biological, digital, or otherwise.

This result contributes to Smale's eighteenth problem [25], which asks: *What are the limits of intelligence, and what are the limits of AI?* We contribute a rate-theoretic answer: under the stated conditions, the observation-to-action bandwidth of any tractable system is characterized by its position on a throughput curve parameterized by treewidth. A more capable system occupies a higher point on this curve — quantitatively different, not qualitatively different.

Outline. The argument proceeds as follows. We state three axioms (§2) — finite-dimensional state, positive observation model, and causal serial interaction — and show that every such system possesses an effective conditional-independence graph (§2) whose treewidth governs information routing. We derive the system's conditional independence graph (§3), establish the tree decomposition and separator structure (§4), define coordinate capacity and information rates (§5), record the Fisher information geometry (§6), and prove the main throughput rate bound (§7). We then establish that tractable inference necessitates bounded treewidth (§9) by four routes: Pitman–Koopman–Darmois for exact inference, Bernstein–von Mises for regular approximate inference under physical non-degeneracy and local prior support, Watanabe's backward reduction for singular latent-variable models, and a comparative posterior-odds reduction for the fully observed residual case. All four routes converge at tractable evaluation of the log-partition function, which requires bounded treewidth under standard complexity assumptions. We further show that the environment's coupling structure determines a lower bound on achievable treewidth (§10), define the effective throughput and model mismatch tax (§11), connect to data-rate control theory (§12), recover the discrete per-step bound as a corollary (§13), discuss related work (§14), and address the result's broader significance (§15).

Complexity-theoretic assumptions. The throughput bound itself (Theorem 7.2) is unconditional. The necessity corollaries additionally invoke two standard conjectures: $\mathrm{FPT} \neq \#\mathrm{W}[1]$ for the exact-inference route, and the Exponential Time Hypothesis (ETH) for the approximate, singular, and comparative routes. Both are stated explicitly where used.

Class-level necessity. The necessity corollaries are stated at the *class level*: they apply to a set $\mathcal{S}$ of continuous systems (a `ContinuousSystemClass` in Lean), and conclude $\exists k,\ \forall\, sys \in \mathcal{S},\ \mathrm{tw}(\mathcal{G}_{\mathrm{eff}}) \leq k$. Since each system bundles its own N, systems in the class may have different numbers of coordinates. The uniform bound k across all sizes is therefore non-trivial.

Formalization posture. The current Lean project is an honest axiomatization. The graph-theoretic layer is partially internalized: tree decompositions, boundary sets, and the boundary cardinality bound are proved directly. The separator theorem, the data processing inequality, the entropy inequalities, Pitman–Koopman–Darmois, Bernstein–von Mises, the Watanabe backward reduction, the comparative-revision backward reduction, and the partition-function hardness statements are imported as explicit axioms. The throughput theorem and the four necessity corollaries are then proved in Lean by composition.

Companion paper. A companion paper [12] establishes a discrete, language-theoretic analogue: any recursively enumerable class of embodied systems supporting uniform tractable belief revision over arbitrary constraint relations with bounded arity has behavior in a $(k{+}1)$-multiple context-free language, where k is the uniform treewidth bound. The discrete result and the continuous result share the same structural bottleneck — a separator of size $k{+}1$ through which all observation-to-action information must pass — but express it in different mathematical languages: a formal language hierarchy (discrete) and an information rate (continuous). Both are accompanied by Lean 4 formalizations.

2 Axioms

Axiom 1 (Finite-dimensional state)**.** The system's state $\theta(t)$ lies in a product of N coordinate spaces:

$$\theta(t) \in \Theta_1 \times \Theta_2 \times \cdots \times \Theta_N,$$

where $N < \infty$ and each Θ_i is a measurable space equipped with a reference measure μ_i.

Remark 2.1. Axiom 1 does not assume discrete domains, finite cardinality, or any algebraic structure on the coordinate spaces. Each Θ_i may be a finite set (with counting measure), a bounded subset of $\mathbb{R}^m$ (with Lebesgue measure), a compact manifold, or any other measurable space. What is required is that the state decomposes into finitely many individually addressable coordinates. Any physical storage medium — digital registers, analog circuits, synaptic weights, molecular concentrations — has finitely many effective degrees of freedom, each with finite precision.

Axiom 2 (Positive observation model)**.** The system maintains a family of observation densities $q(o \mid \theta)$ with respect to a reference measure on the observation space $\mathcal{O}$. For each o, the function $\theta \mapsto q(o \mid \theta)$ is strictly positive as a density on $\Theta_1 \times \cdots \times \Theta_N$ with respect to the product reference measure $\mu_1 \otimes \cdots \otimes \mu_N$:

$$q(o \mid \theta) > 0 \quad \text{for } (\mu_1 \otimes \cdots \otimes \mu_N)\text{-almost all } \theta.$$

Remark 2.2. Positivity means the system can always be surprised: no state assigns zero probability to any observation. This is the regularity condition that ensures the Markov properties (pairwise, local, global) are equivalent. In the finite-domain special case, this is exactly the condition required by the Hammersley–Clifford theorem.

Remark 2.3 (The dual role of positivity). Positivity (Axiom 2) serves two distinct roles in this paper:

- (i) *Combinatorial:* Positivity ensures the equivalence of the Markov properties (Theorem 3.2), promoting pairwise CI to the global Markov property needed for the separator-based throughput bound.
- (ii) *Geometric:* Positivity guarantees common support — the regularity condition required by the Pitman–Koopman–Darmois theorem (Theorem 9.1) — ensuring that the system's effective model belongs to an exponential family whose log-partition function governs the Fisher metric and the tractability constraint.

These are not independent roles. The Hammersley–Clifford factorization (clique potentials on a graph) *is* the exponential family factorization (sufficient statistics indexed by cliques, natural parameters, log-partition function). The graph that determines the CI structure simultaneously determines the sparsity of the Fisher metric and the computational structure of the log-partition function $A(\eta)$. Positivity is the single condition that makes the entire apparatus — CI graph, metric, partition function, tractability — cohere.

Axiom 3 (Causal serial interaction)**.** The system interacts with an environment through a continuous loop:

$$o(t) \longrightarrow \theta(t) \longrightarrow a(t) \qquad \text{for all } t \in \mathbb{R}_{\geq 0}.$$

Actuation is a measurable function of the state: $a(t) = \pi(\theta(t))$. All information from observations that reaches actuation passes through θ.

The state process $\{\theta(t)\}$ is *causal*: for any $s < t$,

$$\theta(t) \perp o_{[0,s)} \mid \theta(s), o_{[s,t]}.$$

That is, the state at time t, given the state at time s and the observations in $[s, t]$, is conditionally independent of all earlier observations. The state is a sufficient statistic for the past.

Remark 2.4. No notion of "discrete step" is required. The serial axiom says that at every instant, the action $a(t)$ is determined by $\theta(t)$, and $\theta(t)$ mediates all information from the observation process $o(\cdot)$ to the action process $a(\cdot)$. The discrete loop $o_t \to \theta_t \to a_t \to o_{t+1} \to \cdots$ is a special case obtained by restricting t to the integers.

The causal Markov condition is the continuous-time analogue of "the state is a sufficient statistic for the observation history." It holds automatically for any system whose state update at time t depends only on $\theta(s)$ and $o_{[s,t]}$ (i.e., does not access $o_{[0,s)}$ directly), which is the standard definition of a causal processor.

Definition 2.5 (Information routing)**.** An undirected graph $\mathcal{G}$ on vertex set $\{1,\dots,N\}$ is an *information-routing graph* for the system if, for any separator C in $\mathcal{G}$ between vertex sets A and B,

$$\theta_A(t) \perp \theta_B(t) \mid \theta_C(t), \theta(s) \qquad \text{for all } s < t$$

with respect to information contributed by observations $o_{[s,t]}$. In particular, let $V_O \subseteq \{1,\dots,N\}$ denote the *observation-coupled coordinates* (those whose update can depend directly on the observation $o(t)$) and $V_A \subseteq \{1,\dots,N\}$ the *actuation-determining coordinates* (those read by the policy π). For any separator C in $\mathcal{G}$ between $V_O \setminus C$ and $V_A \setminus C$:

$$\theta_{V_A \setminus C}(t) \perp o(t) \mid \theta_C(t), \theta(s) \quad \text{for all } s < t.$$

Definition 2.6 (Effective CI graph)**.** The *effective CI graph* $\mathcal{G}_{\text{eff}}$ of a system is the sparsest information-routing graph (Definition 2.5): the sparsest undirected graph on the coordinate indices $\{1,\dots,N\}$ such that information from observations to coordinates separated by a graph separator flows only through that separator.

Remark 2.7 (Effective CI graph is always well-defined)**.** Every system has an effective CI graph: the complete graph K_N is trivially an information-routing graph (since no non-trivial separator exists). The effective CI graph is the *sparsest* such graph, and its treewidth determines the tightest bound.

Theorem 2.8 (CI graph identification for faithful inference)**.** *Under Axioms 1–3:*

(a) Exact Bayesian inference with factored prior: *The effective CI graph equals the conditional independence graph $\mathcal{G}(q)$ (Definition 3.1). This is a theorem: the posterior is a Markov random field on $\mathcal{G}(q)$, and the global Markov property — guaranteed by positivity (Axiom 2) — gives the separator property as a consequence [16].*

(b) Graph-respecting approximate inference *(loopy belief propagation, variational message passing, expectation propagation, junction tree algorithms): The effective CI graph is a subgraph of the algorithm's computation graph. Messages are routed along edges; every computational path from an observation-coupled node to a separated node must cross the separator. The values may be approximate; the routing is exact.*

(c) Dense / amortized inference *(fully connected networks, transformers with global attention): The effective CI graph may be as dense as K_N. The bound applies to its treewidth, which may be $N-1$ (vacuously large).*

Proof. (a) By Axiom 2 and Theorem 3.2, the model $q(o \mid \theta)$ satisfies the global Markov property on $\mathcal{G}(q)$. The posterior $p(\theta \mid o)$ inherits the Markov property (it is proportional to $q(o \mid \theta) \cdot p(\theta)$ with $p(\theta)$ factored on the same graph). The global Markov property gives: for any separator C between A and B in $\mathcal{G}(q)$, $\theta_A \perp \theta_B \mid \theta_C$ under the posterior. Since Bayesian update computes the posterior exactly, new observation information reaching θ_A through the posterior cannot bypass C. Hence $\mathcal{G}(q)$ is an information-routing graph (Definition 2.5), and since $\mathcal{G}(q)$ is the graph encoding pairwise conditional independences (Definition 3.1), no sparser graph satisfies the routing property. So $\mathcal{G}_{\text{eff}} = \mathcal{G}(q)$.

(b) Graph-respecting algorithms route messages along a fixed computation graph $\mathcal{G}_{\text{comp}}$. Every information path from an observation-coupled coordinate to any other coordinate traverses only edges of $\mathcal{G}_{\text{comp}}$. Hence $\mathcal{G}_{\text{comp}}$ is an information-routing graph, so $\mathcal{G}_{\text{eff}} \subseteq \mathcal{G}_{\text{comp}}$.

(c) Any system has $\mathcal{G}_{\text{eff}} = K_N$ (trivially). □

Remark 2.9 (Information routing is derived, not assumed). Definition 2.5 specifies what it means for a graph to route information for the system. Every system possesses at least one such graph (the complete graph K_N), so the effective CI graph is always well-defined. Theorem 2.8 identifies it in the main cases of interest: for exact Bayesian inference it is the conditional-independence graph $\mathcal{G}(q)$; for graph-respecting approximate inference it is a subgraph of the computation graph; and in the dense extreme it is K_N. Information routing through separators is thus a *consequence* of the axioms and the inference method, not a separate assumption. The throughput bound is governed by the treewidth of whichever of these graphs the system realizes.

3 The Conditional Independence Graph

Definition 3.1 (Conditional independence graph). The *conditional independence graph* $\mathcal{G}(q)$ is an undirected graph on vertex set $\{1, \ldots, N\}$ defined by:

$$(i, j) \notin \mathcal{G}(q) \iff \theta_i \perp\!\!\!\perp \theta_j \mid \theta_{\{1,\ldots,N\}\setminus\{i,j\}} \quad \text{under } q(o \mid \theta) \text{ for all } o \in \mathcal{O}.$$

Equivalently: i and j are adjacent if and only if there exists an observation o under which θ_i and θ_j are not conditionally independent given all other coordinates.

Theorem 3.2 (Markov property equivalence). *Under Axiom 2 (positivity), the following are equivalent for the observation model $q(o \mid \theta)$ on the product space $\Theta_1 \times \cdots \times \Theta_N$:*

(a) Pairwise Markov property: *non-adjacent coordinates are conditionally independent given all others.*

(b) Global Markov property: *for any triple (A, B, C) with C separating A from B in $\mathcal{G}(q)$, $\theta_A \perp\!\!\!\perp \theta_B \mid \theta_C$.*

(c) Factorization: $q(o \mid \theta) = \prod_{c \in \text{cliques}(\mathcal{G}(q))} \phi_c(\theta_c, o)$ *for appropriate clique potentials* $\phi_c > 0$.

Proof. The equivalence $(a) \Leftrightarrow (b) \Leftrightarrow (c)$ under positivity is proved in Lauritzen [16], Chapter 3, for general product spaces with densities positive with respect to a product reference measure. The classical Hammersley–Clifford theorem [8] establishes $(a) \Leftrightarrow (c)$ for the special case of finite sets with counting measure. □

Remark 3.3 (What positivity buys). Without positivity, the pairwise Markov property is strictly weaker than the global Markov property. Positivity closes the gap: pairwise CI (which defines $\mathcal{G}(q)$) is promoted to global CI (which gives the separator property needed in the main proof). This is the only role of positivity in the throughput bound (Theorem 7.2). Positivity serves a second role in the necessity argument (§9): it provides the common-support condition required by the Pitman–Koopman–Darmois theorem to derive exponential family structure from the axioms (Theorem 9.1).

Remark 3.4 (Edges encode coupling). An edge $(i, j) \in \mathcal{G}(q)$ means: there exists an observation under which knowing θ_i changes what can be inferred about θ_j, even conditioned on all other coordinates. No edge means the two coordinates are informationally decoupled across all observations and can always be updated independently.

Remark 3.5 (Coordinate dependence). $\mathcal{G}(q)$ is defined relative to the system's actual coordinates. A reparameterization could yield a denser graph. The system's coordinates are the relevant ones because they determine what the inference computation operates on.

4 Tree Decomposition, Treewidth, and Separators

Definition 4.1 (Tree decomposition). A *tree decomposition* of a finite simple graph $G = (V, E)$ is a pair $(T, \{B_t\}_{t \in V(T)})$ where T is a tree and each bag $B_t \subseteq V$ satisfies:
(T1) **Vertex cover:** $\bigcup_{t \in V(T)} B_t = V$.
(T2) **Edge cover:** For every edge $\{u, v\} \in E$, there exists $t \in V(T)$ with $u, v \in B_t$.
(T3) **Running intersection:** For each vertex $x \in V$, the set $\{t \in V(T) : x \in B_t\}$ induces a connected subtree of T.
The *width* of the decomposition is $\max_t |B_t| - 1$.

Definition 4.2 (Treewidth). The *treewidth* of a graph G, denoted $\mathrm{tw}(G)$, is the minimum width over all tree decompositions of G.

Definition 4.3 (Separator). Let $A, B, C \subseteq V$. We say that C is a *separator* between A and B if every walk from a vertex of $A \setminus C$ to a vertex of $B \setminus C$ visits some vertex of C.

Fix a rooted tree decomposition $(T, \{B_t\})$ and a node t. Let $V_t^{\downarrow}$ be the union of the bags in the subtree rooted at t, and let

$$\partial_t = V_t^{\downarrow} \cap \bigcup \{B_s : s \notin \text{subtree at } t\}$$

be the boundary of that subtree.

Lemma 4.4 (Boundary in bag). *For every node t, one has $\partial_t \subseteq B_t$.*

Proof. Let $v \in \partial_t$. Then v occurs in some bag below t and in some bag outside the subtree at t. By condition (T3), the bags containing v form a connected subtree of T. That connected subtree must contain the unique path between those two occurrences, hence it contains t. Therefore $v \in B_t$. □

Lemma 4.5 (Boundary cardinality bound). *If a tree decomposition has width at most k, then $|\partial_t| \leq k + 1$ for every node t.*

Proof. By Lemma 4.4, $\partial_t \subseteq B_t$. Since the decomposition has width at most k, every bag has size at most $k + 1$. Hence $|\partial_t| \leq k + 1$. □

Theorem 4.6 (Separator property). *Let $G = (V, E)$ with $\mathrm{tw}(G) \leq k$. For any two disjoint non-empty vertex sets $A, B \subseteq V$, there exists a set $C \subseteq V$ with $|C| \leq k+1$ such that every path in G from a vertex in $A \setminus C$ to a vertex in $B \setminus C$ passes through C.*

Proof. Let $(T, \{B_t\})$ be a tree decomposition of G of width k. Root T at an arbitrary node. Since A and B are non-empty, there exist bags B_s and B_t with $B_s \cap A \neq \emptyset$ and $B_t \cap B \neq \emptyset$. On the unique path from s to t in T, let $e = (u, v)$ be any edge. The set $C = B_u \cap B_v$ satisfies $|C| \leq k + 1$ (since $|B_u|, |B_v| \leq k + 1$). Removing the edge e from T partitions V into two sets V_u, V_v (vertices appearing only in bags on the u-side or v-side respectively), with $C = V_u \cap V_v$. By condition (T3), any edge $(x, y) \in E$ with $x \in V_u \setminus C$ and $y \in V_v \setminus C$ would require a bag containing both x and y, but x appears only in u-side bags and y only in v-side bags (outside C), a contradiction. Hence every path from $A \setminus C$ to $B \setminus C$ passes through C. □

5 Coordinates and Capacity

Definition 5.1 (Coordinate capacity). The *capacity* of coordinate i is:

$$C_i = \sup_{\rho_i} \; H_{\mu_i}(\rho_i),$$

where the supremum is over all distributions ρ_i on Θ_i that the system can physically realize, and H_{μ_i} denotes differential entropy with respect to the reference measure μ_i.

Remark 5.2 (Finiteness of capacity). $C_i < \infty$ for every physical implementation:

(i) *Digital coordinates:* Θ_i is a finite set, μ_i is counting measure, and $C_i = \log|\Theta_i|$.

(ii) *Analog coordinates with bounded range and finite precision:* $\Theta_i \subseteq \mathbb{R}^m$ bounded, with precision $\epsilon_i > 0$, so $C_i \le \log(\mathrm{vol}(\Theta_i)/\epsilon_i^m)$.

(iii) *Variance-constrained coordinates:* $C_i = \frac{1}{2}\log(2\pi e\,\sigma_i^2)$ (Gaussian maximizes entropy under variance constraint).

Finite capacity is a consequence of physical realizability, not a mathematical axiom imposed on the formalism.

Definition 5.3 (Coordinate information rate). The *information rate* of coordinate i is the conditional entropy rate:

$$R_i = \limsup_{\Delta t \to 0^+} \frac{1}{\Delta t} H\big(\theta_i(t+\Delta t) \mid \theta(t)\big),$$

the rate at which the state of coordinate i changes, measured in bits per unit time [22].

Remark 5.4 (Rate is finite). $R_i < \infty$ for any physical system. Specific realizations include:

(i) *Digital systems clocked at frequency f:* $R_i = f \cdot \log|\Theta_i|$ bits per second.

(ii) *Analog channels with bandwidth B_i and signal-to-noise ratio S_i:* $R_i \le B_i \log(1+S_i)$ (Shannon channel capacity).

(iii) *Neural substrates:* R_i bounded by synaptic transmission rate $\times$ bits per event.

Finiteness of R_i follows from finite bandwidth and finite signal power — physical constraints, not mathematical discretization.

Definition 5.5 (Maximum capacity and rate).

$$C_{\max} = \max_{1 \le i \le N} C_i, \qquad R_{\max} = \max_{1 \le i \le N} R_i.$$

6 Fisher Information (Supplementary)

When the coordinate spaces are subsets of $\mathbb{R}^m$ and q is differentiable in quadratic mean (DQM), the Fisher information defines a Riemannian metric on the state space.

Definition 6.1 (Fisher information matrix).

$$G_{ij}(\theta) = \mathbb{E}_{o \sim q(\cdot|\theta)}\left[\frac{\partial \log q(o \mid \theta)}{\partial \theta_i} \cdot \frac{\partial \log q(o \mid \theta)}{\partial \theta_j}\right].$$

Remark 6.2 (Relation to CI graph). For exponential families, $G_{ij}(\theta) = \mathrm{Cov}_q[T_i(o), T_j(o)]$ and the sparsity pattern of G coincides exactly with $\mathcal{G}(q)$. For general smooth families, G may be denser: score functions can be correlated through o even when θ_i and θ_j share no clique. The CI graph remains the primary structural object; the Fisher matrix provides quantitative geometry but may have strictly denser support.

7 The Throughput Rate Bound

Definition 7.1 (Instantaneous information throughput). The *instantaneous information throughput* at time t is:

$$\mathcal{I}(t) = \limsup_{\Delta t \to 0^+} \frac{1}{\Delta t} I\big(o_{[t,\,t+\Delta t]};\ a_{[t,\,t+\Delta t]} \mid \theta(t)\big),$$

the rate at which information from the observation process reaches the action process, conditioned on the current state.

Theorem 7.2 (Information throughput rate bound)**.** *Let a system satisfy Axioms 1–3, and let $\mathcal{G}_{\text{eff}}$ be its effective CI graph (Definition 2.6) with $k = \text{tw}(\mathcal{G}_{\text{eff}})$. Then:*

$$\mathcal{I}(t) \;\leq\; (k+1) \cdot R_{\max}$$

at every time t, for every observation process.

Proof. The proof proceeds in five steps, applied at an arbitrary instant t.

Step 1: Derived structure. By Axiom 1, $\theta(t) = (\theta_1(t), \ldots, \theta_N(t)) \in \Theta_1 \times \cdots \times \Theta_N$ with $N < \infty$. By Axiom 2 and Theorem 3.2, the observation model has a conditional independence graph $\mathcal{G}(q)$ on N vertices, and positivity guarantees that the global Markov property holds. By Theorem 2.8, the effective CI graph $\mathcal{G}_{\text{eff}}$ is well-defined; for exact Bayesian inference, $\mathcal{G}_{\text{eff}} = \mathcal{G}(q)$.

Step 2: Separator. $\mathcal{G}_{\text{eff}}$ has treewidth k. By Theorem 4.6, there exists $C \subseteq \{1, \ldots, N\}$ with $|C| \leq k+1$ separating $V_O \setminus C$ from $V_A \setminus C$ in $\mathcal{G}_{\text{eff}}$.

Step 3: Information routing. By the definition of $\mathcal{G}_{\text{eff}}$ (Definition 2.5), $\theta_{V_A \setminus C}(t')$ depends on $o_{[t,t']}$ only through $\theta_C(t')$ (given $\theta(t)$) for any $t' > t$. By Axiom 3, $a(t') = \pi(\theta(t'))$ is a measurable function of the full state, so $a(t')$ is determined by $(\theta_C(t'),\, \theta(t))$.

Step 4: Data processing inequality. Since $a(t')$ is a function of $(\theta_C(t'),\, \theta(t))$, the data processing inequality [2] gives, for any $\Delta t > 0$:

$$I\big(o_{[t,t+\Delta t]};\; a_{[t,t+\Delta t]} \;\big|\; \theta(t)\big) \;\leq\; I\big(o_{[t,t+\Delta t]};\; \theta_C(t{+}\Delta t) \;\big|\; \theta(t)\big).$$

All information from observations reaching actions must pass through the separator coordinates.

Step 5: Rate bound on separator.

$$\begin{aligned}
\frac{1}{\Delta t}\, I\big(o_{[t,t+\Delta t]};\; \theta_C(t{+}\Delta t) \;\big|\; \theta(t)\big) &\leq \frac{1}{\Delta t}\, H\big(\theta_C(t{+}\Delta t) \;\big|\; \theta(t)\big) && (1)\\
&\leq \frac{1}{\Delta t} \sum_{c \in C} H\big(\theta_c(t{+}\Delta t) \;\big|\; \theta(t)\big) && (2)\\
&\leq \sum_{c \in C} R_c && (3)\\
&\leq (k+1) \cdot R_{\max}. && (4)
\end{aligned}$$

Here: (1) is $I(X;Y \mid Z) \leq H(Y \mid Z)$ (mutual information does not exceed entropy); (2) is subadditivity of entropy; (3) follows from taking $\limsup_{\Delta t \to 0^+}$ and applying Definition 5.3; and (4) uses $|C| \leq k+1$ and the definition of $R_{\max}$. $\square$

Corollary 7.3 (Integral form)**.** *For any interval $[s,t] \subseteq \mathbb{R}_{\geq 0}$:*

$$I\big(o_{[s,t]};\; a_{[s,t]} \;\big|\; \theta(s)\big) \;\leq\; (k+1) \int_s^t R_{\max}(\tau)\, d\tau.$$

When $R_{\max}$ is constant: $I(o_{[s,t]};\, a_{[s,t]} \mid \theta(s)) \leq (k+1) \cdot R_{\max} \cdot (t-s)$.

Proof. Let $s = t_0 < t_1 < \cdots < t_n = t$ be a partition of $[s,t]$. By Axiom 3 (actions determined by state):

$$I\big(o_{[s,t]};\; a_{[s,t]} \;\big|\; \theta(s)\big) \;\leq\; I\big(o_{[s,t]};\; \theta(t_1), \ldots, \theta(t_n) \;\big|\; \theta(s)\big).$$

By the chain rule for mutual information:

$$= \sum_{i=1}^{n} I\big(o_{[s,t]};\; \theta(t_i) \;\big|\; \theta(s),\, \theta(t_1), \ldots, \theta(t_{i-1})\big).$$

By the causal Markov property (Axiom 3), $\theta(t_i)$ given $\theta(t_{i-1})$ depends on $o_{[s,t]}$ only through $o_{[t_{i-1},t_i]}$, so each summand satisfies:

$$I\big(o_{[s,t]};\ \theta(t_i)\ \big|\ \theta(s),\, \theta(t_1), \ldots, \theta(t_{i-1})\big)\ \le\ I\big(o_{[t_{i-1},t_i]};\ \theta(t_i)\ \big|\ \theta(t_{i-1})\big).$$

Applying Theorem 7.2 to each subinterval:

$$I\big(o_{[t_{i-1},t_i]};\ \theta(t_i)\ \big|\ \theta(t_{i-1})\big)\ \le\ (k{+}1) \cdot R_{\max} \cdot (t_i - t_{i-1}).$$

Summing over the partition:

$$I\big(o_{[s,t]};\ a_{[s,t]}\ \big|\ \theta(s)\big)\ \le\ (k{+}1) \cdot R_{\max} \cdot \sum_{i=1}^{n} (t_i - t_{i-1}) = (k{+}1) \cdot R_{\max} \cdot (t - s).$$

For time-varying $R_{\max}(\tau)$, the same argument yields the integral form: each subinterval contributes at most $(k{+}1) \cdot R_{\max}(\tau_i^*) \cdot (t_i - t_{i-1})$ for some $\tau_i^* \in [t_{i-1}, t_i]$, and the sum is a Riemann sum converging to $(k{+}1) \int_s^t R_{\max}(\tau)\, d\tau$ as the partition refines, since $R_{\max}(\tau) < \infty$ is bounded (Remark 5.4) and hence Riemann integrable on compact intervals. □

Remark 7.4 (The bound is structural). The bound holds for *every* observation process. The bottleneck is the system's computational architecture: information from observations reaches actuation-determining coordinates only through the separator. This is a property of how the system routes information, not of the observation statistics.

Remark 7.5 (Unconditionality). Theorem 7.2 is unconditional. No complexity-theoretic hypothesis enters its proof. The bound follows from the data processing inequality, which is a mathematical identity — not a conjecture. No algorithm, exact or approximate, variational or neural, can create mutual information between $o(t)$ and $a(t)$ that exceeds the information rate through the separator. This is analogous to conservation laws in physics: a thermodynamic bound, not a computational one.

Remark 7.6 (Model mismatch). When the true distribution p differs from the system's model q, the system still admits at most $(k{+}1) \cdot R_{\max}$ bits per second of observation-to-action throughput, but those bits may be poorly calibrated. The bound constrains the *quantity* of information flowing from observations to actions; model mismatch degrades the *quality*. The gap between raw and effective throughput is formalized as the *mismatch tax* in §11.

8 Treewidth as a System Parameter

Theorem 7.2 bounds throughput in terms of $k = \mathrm{tw}(\mathcal{G}_{\text{eff}})$. The treewidth is a property of the system's inference architecture — determined by its effective CI graph. To achieve higher throughput, a system needs an effective CI graph of higher treewidth, which requires maintaining more complex dependency structure in its internal state.

Remark 8.1 (Reading off treewidth). For a specific system, $\mathcal{G}_{\text{eff}}$ can often be read off the model architecture and its treewidth computed directly. Theorem 7.2 then gives a throughput rate bound for that specific system without any further assumptions. When the state evolves by coupled dynamics $\dot{\theta}_i = f_i(\theta_{\text{neighbors}(i)}, o)$, the sparsity pattern of the coupling directly yields $\mathcal{G}_{\text{eff}}$. Neural ODEs, recurrent dynamics, attractor networks, reservoir computers — in each case, the treewidth is determined by the sparsity of the coupling, and the rate bound follows immediately.

Remark 8.2 (Resource cost of higher treewidth). Higher treewidth is not free. A system with $\mathrm{tw}(\mathcal{G}_{\text{eff}}) = k$ must jointly process cliques of size up to $k{+}1$ during inference. The time and memory cost of exact inference scales as $O(N \cdot \exp(k))$ [17]. Thus the treewidth k governs both the information-theoretic ceiling (throughput rate) and the computational cost (inference time). A system that wants more throughput must pay for it in computational resources — there is no free lunch.

Remark 8.3 (Diminishing returns from scaling). The throughput bound grows linearly in k, but the computational cost of inference grows exponentially: $O(N \cdot d^{k+1})$ for effective dimensionality d [17]. A system with compute budget B operations per second can sustain treewidth at most $k_{\max} = O(\log_d(B/N))$, yielding throughput rate $O(\log(B) \cdot R_{\max})$ — sub-linear in hardware resources. Doubling compute adds at most one unit of treewidth and $R_{\max}$ bits per second of throughput. Gustafson's law does not rescue this scaling: Gustafson requires the parallelizable fraction to grow with problem size, but the CALM theorem [9] tells us that the serial fraction of inference (the junction tree message schedule) is non-vanishing because belief revision is non-monotonic. The scaling regime is Amdahl [1], not Gustafson: more hardware helps, but with exponentially diminishing returns per unit of treewidth gained.

9 Tractable Inference Necessitates Bounded Treewidth

Theorem 7.2 bounds throughput for any given treewidth k. This section establishes that k is not a free parameter: any system performing tractable inference must have $k = O(1)$.

Here "tractable inference" means polynomial-time computation with bounded approximation error — not merely polynomial runtime regardless of output quality. A system running mean-field or loopy BP in $O(N)$ time on a high-treewidth graph executes quickly but produces posteriors whose approximation error is uncontrolled. Such a system is rate-bounded by Theorem 7.2: it cannot route more than $(k+1) \cdot R_{\max}$ bits per second of observation-to-action information regardless of how fast it runs, and the information it does route may be poorly calibrated. The necessity result below shows that achieving bounded approximation error in polynomial time forces $k = O(1)$.

The argument proceeds via four routes, all converging at the partition function:

(i) **Exact inference** (§9.1): The axioms alone imply exponential family structure via Pitman–Koopman–Darmois.

(ii) **Approximate inference** (§9.2): Under a physical non-degeneracy condition and a mild local-prior-support assumption, the Bernstein–von Mises theorem establishes that any system satisfying the axioms converges asymptotically into exponential family territory. This is the load-bearing bridge that extends necessity from exact to approximate inference — without it, the result would apply only to exact Bayesian systems, excluding every real implementation.

(iii) **Singular models** (§9.3): For models with degenerate Fisher information and latent variables, a backward reduction via Watanabe's uniqueness theorem shows that any polynomial-time learner achieving the optimal generalization rate must implicitly compute the partition function.

(iv) **Fully observed models** (§9.4): For fully observed singular models, tractable comparative Bayesian revision (posterior-odds computation) with a partition anchor reduces to the partition function.

Routes (i)–(ii) derive exponential family structure from the axioms; routes (iii)–(iv) bypass the exponential family step entirely and backward-reduce directly to the partition function. In all four cases, **partition function tractability** (§9.5): tractable evaluation of the log-partition function — exactly or approximately — requires bounded treewidth, conditional on standard complexity assumptions.

9.1 The axioms entail exponential family structure

Theorem 9.1 (Finite sufficiency implies exponential family)**.** *Let a system satisfy Axioms 1–3 and perform exact inference (the state $\theta(t)$ is a sufficient statistic for the observation history in the Fisher–Neyman sense). Then the observation model $q(o \mid \theta)$ belongs to an exponential family with sufficient statistics of dimension at most N.*

Proof. By Axiom 3, the state $\theta(t)$ is a sufficient statistic for the observation history $o_{[0,t]}$: the posterior $p(\theta \mid o_{[0,t]})$ depends on the observations only through $\theta(t)$. By Axiom 1, $\theta(t)$ has fixed dimension N, independent of how many observations have been processed. By Axiom 2, the observation model $q(o \mid \theta)$ has support independent of θ (strict positivity implies common support). By the Pitman–Koopman–Darmois theorem [13], a family of distributions with common support that admits a sufficient statistic of fixed finite dimension is necessarily an exponential family.

Note on i.i.d. vs. sequential observations. The classical PKD theorem is stated for i.i.d. samples from a fixed distribution. Here, the observations $o_{[0,t]}$ form a sequential process, not an i.i.d. sample. The PKD argument applies because it constrains the *model family* $q(o \mid \theta)$, not the data stream: the fact that $\theta(t)$ is N-dimensional and sufficient for all observation histories means the model must admit N-dimensional sufficient statistics, and PKD forces exponential family structure on any such model. The conclusion is about the parametric form of q, not the distributional properties of the observation sequence. □

Remark 9.2 (PKD conditions match the axioms). The Pitman–Koopman–Darmois theorem requires three conditions: (i) the family is dominated by a common σ-finite measure (guaranteed by the reference measure in Axiom 1); (ii) the support does not depend on the parameter (guaranteed by positivity, Axiom 2); (iii) the sufficient statistic has fixed finite dimension (guaranteed by finite-dimensional state, Axiom 1, combined with sufficiency from Axiom 3). These are exactly the conditions the axioms establish. Accordingly, any system performing exact inference under the axioms lies in an exponential-family regime.

9.2 Approximate inference: asymptotic exponential family structure

When a system performs approximate inference, the state $\theta(t)$ is a lossy summary of the observation history — not a sufficient statistic in the Fisher–Neyman sense. The Pitman–Koopman–Darmois theorem (Theorem 9.1) does not apply: its hypothesis (exact sufficiency) fails. The throughput bound (Theorem 7.2) holds unconditionally — the DPI constrains lossy processing at least as tightly as lossless processing — but we need a different route to establish that *bounded treewidth is necessary* for approximate inference systems.

The Bernstein–von Mises theorem provides this route. It is the load-bearing result for all practical systems: without it, the necessity argument would apply only to exact Bayesian inference, a mathematical idealization that no physical system achieves.

Condition 9.3 (Physical non-degeneracy)**.** The observation model $q(o \mid \theta)$ satisfies:

(i) *Smoothness:* The observation model is *differentiable in quadratic mean* (DQM) at the true parameter θ_0: $\int [\sqrt{q(o \mid \theta_0 + h)} - \sqrt{q(o \mid \theta_0)} - \frac{1}{2} h^\top \dot{\ell}(o \mid \theta_0) \sqrt{q(o \mid \theta_0)}]^2 \, d\mu(o) = o(|h|^2)$.

(ii) *Non-degeneracy:* The Fisher information matrix $G(\theta_0)$ (Definition 6.1) is positive definite.

Condition 9.4 (Local prior support)**.** The system's prior density $\pi(\theta)$ is continuous and strictly positive in a neighborhood of the true parameter θ_0.

Remark 9.5 (Why prior support is mild). This is not a smoothness condition on the likelihood. It is an expressivity condition on the system's beliefs: before seeing data, the system has not assigned zero weight to the truth. A prior that excludes the true parameter cannot be repaired by Bayesian updating. For a system intended to support general-purpose belief revision, local prior support is therefore the minimal requirement that the true environment state is not ruled out in advance.

Remark 9.6 (The non-degeneracy condition excludes only pathological cases). *Smoothness* fails only for observation models with discontinuous log-likelihoods — distributions whose support depends on the parameter, such as Uniform$(0, \theta)$. But Axiom 2 already excludes θ-dependent support: strict positivity guarantees common support for all θ. Any positive observation model with continuous densities satisfies DQM generically. Physical sensors produce observations corrupted by thermal, shot, or quantization noise, all of which yield DQM likelihoods. DQM is strictly weaker than C^2 smoothness: it accommodates non-smooth models (e.g., LASSO, ReLU networks) provided the square-root density is locally linearizable.

Non-degeneracy fails when distinct parameter values produce identical observation distributions — meaning some coordinates of θ carry zero information and are operationally inert. These inert coordinates can be removed without affecting any inference quantity; a positive definite Fisher matrix simply asserts that every coordinate of θ does observable work. No physical system maintains state variables with zero observable consequences.

Remark 9.7 (Overparameterized models and singular Fisher matrices). Highly overparameterized models—deep neural networks, for instance—frequently have singular (rank-deficient) Fisher information matrices because many distinct weight configurations produce the same input–output map. This does *not* invalidate the necessity argument. Parameter redundancy is an artifact of the parameterization, not of the underlying function class. Quotienting the parameter space by the equivalence relation "$\theta \sim \theta'$ iff $q(\cdot \mid \theta) = q(\cdot \mid \theta')$" recovers the model's *identifiable manifold*: the space of genuinely distinct distributions the architecture can represent. The dimension of this manifold equals the rank of the Fisher matrix, and the reparameterized Fisher on the quotient is non-degenerate by construction. BvM (Theorem 9.8) applies to this identifiable quotient.

The trilemma then follows as before: if the identifiable manifold is expressive enough to represent arbitrary distributions on a dense graph $\mathcal{G}_{\mathrm{eff}}$, the BvM limit forces it into a dense exponential family whose partition function is ETH-hard to evaluate (Theorem 9.36). Singular parameterizations therefore do not escape the bottleneck—either their identifiable core restricts representation (and treewidth remains bounded), or inference is intractable.

Theorem 9.8 (Bernstein–von Mises: asymptotic exponential family structure). *Under Axioms 1–3 and Conditions 9.3 and 9.4, as the effective sample size $n \to \infty$:*

$$p(\theta \mid o_1, \ldots, o_n) \xrightarrow{d} \mathcal{N}\big(\hat{\theta}_{\mathrm{MLE}},\ \ n^{-1}\, G(\theta_0)^{-1}\big).$$

The posterior concentrates into a Gaussian — itself an exponential family member — with covariance determined by the inverse Fisher information.

Proof. This is the classical Bernstein–von Mises theorem [26]. The required conditions are: (i) the true parameter θ_0 lies in the interior of Θ (ensured by common support: Axiom 2 guarantees all of Θ is in the support); (ii) the model is identifiable and DQM (Condition 9.3: positive definite $G(\theta_0)$ implies local identifiability; DQM is stated directly); (iii) the prior is positive and continuous at θ_0 (Condition 9.4); (iv) the Fisher information $G(\theta_0)$ is non-degenerate (Condition 9.3). All conditions are satisfied.

Note on i.i.d. vs. sequential observations. The classical BvM theorem is stated for i.i.d. samples. In a filtering setting with non-stationary observations, local versions of BvM apply under mixing or ergodicity conditions that physical systems generically satisfy [26]. What matters is that the posterior concentrates at rate $O(1/n)$ in the effective sample count, and this holds in the regular regime once observations are sufficiently informative, the Fisher information is non-degenerate, and the prior has local support. □

Remark 9.9 (Why BvM is load-bearing, not supplementary). For exact inference, the PKD theorem (Theorem 9.1) establishes exponential family structure directly — no asymptotics needed. But no real system performs exact Bayesian inference. Every physical implementation uses some form of approximation: variational methods, MCMC, particle filters, gradient-based optimization, neural network amortization. For all such systems, the state $\theta(t)$ is a *lossy compression* of the observation history: PKD's hypothesis fails and cannot establish necessity.

The Bernstein–von Mises theorem bridges this gap. It establishes that under the axioms and Conditions 9.3 and 9.4, *every* system's posterior — regardless of the inference algorithm — converges into exponential family territory. The convergence is a property of the statistical model and the data-generating process, not the algorithm. The inference algorithm determines how well the system *tracks* this convergence; the model determines *where* the posterior converges to.

Once converged, the system's effective model is an exponential family on its CI graph $\mathcal{G}_{\text{eff}}$. The partition function tractability theorem (§9.5) then forces $\text{tw}(\mathcal{G}_{\text{eff}}) = O(1)$.

The necessity of bounded treewidth for approximate inference is therefore an *asymptotic* result: it applies once the posterior has concentrated. During the transient regime before convergence, the throughput bound (Theorem 7.2) holds unconditionally, but the treewidth necessity does not yet bind.

Remark 9.10 (The architecture argument). A system's computational graph $\mathcal{G}_{\text{eff}}$ is a *structural commitment*: it is fixed at design time, before the system encounters any specific environment. The parameters on this graph are learned or adapted from data; the graph topology itself is the architectural choice.

For the necessity argument, the crucial quantifier is "for all models on $\mathcal{G}_{\text{eff}}$." A specific model on a high-treewidth graph might happen to be easy to compute (just as a specific SAT instance might be easy despite NP-hardness in general). But the architecture must support tractable inference for *whatever model could arise* — different environments, different observation statistics, different parameter regimes. BvM guarantees that each such model converges to an exponential family member on $\mathcal{G}_{\text{eff}}$, and the family of all exponential families on $\mathcal{G}_{\text{eff}}$ spans all possible clique potentials. If even one member of this family makes inference intractable, the architecture is insufficient.

The "for all models" quantifier is not an artificial strengthening: it reflects the genuine design constraint that an architecture must work across the range of environments it will encounter, not just on a single benign instance.

9.3 Singular models: the backward reduction

The BvM bridge (Theorem 9.8) requires non-degenerate Fisher information (Condition 9.3) and local prior support (Condition 9.4). Singular models — mixtures at component boundaries, neural networks with redundant neurons, reduced-rank regressions, hidden Markov models with redundant states — violate this condition: the Fisher matrix is rank-deficient at the true parameter because distinct weight configurations produce identical distributions. Remark 9.7 argues that quotienting by the identifiability relation recovers non-degeneracy, but the quotient construction is algebraic and may not preserve tractability of inference. We now establish the same bounded-treewidth conclusion for singular models by a different route that bypasses BvM entirely.

Condition 9.11 (Latent variable structure)**.** The system is a graphical model with latent variables: the likelihood of the observed data given parameters w requires marginalizing over hidden variables h:

$$p(x_{\text{obs}} \mid w) = \frac{1}{Z(w)} \sum_h \prod_{C \in \text{cliques}(G)} \psi_C(x_C, h_C; w),$$

where $Z(w) = \sum_{x,h} \prod_C \psi_C$ is the partition function.

Remark 9.12 (Why latent variables are load-bearing). For fully-observed models without latent variables, the likelihood is $p(x \mid w) = \prod_C \psi_C(x_C;\, w)/Z(w)$, and MCMC methods can use likelihood *ratios* $p(x \mid w_1)/p(x \mid w_2)$ in which Z cancels. Such methods achieve tractable inference without computing Z. With latent variables, the marginalization $\sum_h$ is a sum-product computation on the graph whose complexity is governed by treewidth. Likelihood ratios no longer cancel the partition function because the marginalization changes structure at each parameter value. Condition 9.11 is therefore genuinely necessary: it ensures that the partition function cannot be circumvented. In practice, this is a mild condition: virtually all interesting singular models (mixture models, deep networks, HMMs, RBMs, VAEs) have latent variables.

Condition 9.13 (Bayesian computation requires likelihood evaluation)**.** Computing the Bayes posterior predictive

$$p^*(x_{n+1} \mid x^n) = \int p(x_{n+1} \mid w)\, \pi(w \mid x^n)\, dw$$

for a latent-variable graphical model requires evaluating the likelihood $p(x_{\mathrm{obs}} \mid w)$ at polynomially many parameter values w. Equivalently: there is no algorithm that computes the posterior-predictive integral without accessing its integrand.

Remark 9.14 (Status of Condition 9.13). Condition 9.13 is close to a definitional fact: the Bayes predictive *is* an integral over the likelihood, and computing an integral requires evaluating the integrand. Every known method for posterior computation — MCMC, variational inference, sequential Monte Carlo, numerical quadrature — evaluates $p(x_{\mathrm{obs}} \mid w)$ at polynomially many parameter values. We state it as an explicit condition rather than leaving it implicit because, strictly speaking, proving that *no algorithm* can compute the integral without pointwise likelihood evaluations would require an unconditional computational lower bound — and such lower bounds are not available for essentially any nontrivial computational problem. In practice, Condition 9.13 is no more speculative than the Exponential Time Hypothesis: both assert that a natural computational task cannot be solved by tricks that bypass its defining structure.

Theorem 9.15 (Backward reduction: learning success implies partition-function tractability)**.** *Let a system satisfy Conditions 9.11 and 9.13, and suppose a polynomial-time learner $\mathcal{A}$ achieves the Watanabe-optimal generalization rate:*

$$G_n = K(w^*) + \frac{\lambda}{n} \log n + o\!\left(\frac{\log n}{n}\right),$$

where λ is the real log canonical threshold of the model. Then the system's approximate partition function is tractable: $Z(w)$ can be evaluated to polynomial accuracy in polynomial time.

Proof. The argument combines uniqueness (backward) with computational analysis (forward):
Step 1 (Uniqueness). By Watanabe [29, Theorem 6.7], the Bayes posterior predictive

$$p(x_{n+1} \mid x^n) = \frac{Z_{n+1}}{Z_n} = \frac{\int p(x_{n+1} \mid w)\, p(x^n \mid w)\, \varphi(w)\, dw}{\int p(x^n \mid w)\, \varphi(w)\, dw}$$

is the unique predictor achieving the rate $\lambda \log n/n$ for singular models [29, Theorem 7.2]. Since $\mathcal{A}$ achieves this rate, it must compute a quantity equivalent to the Bayes predictive (up to $o(\log n/n)$ terms).
Step 2 (Internal requirements). By Condition 9.13, computing the Bayes predictive requires evaluating the likelihood $p(x_{\mathrm{obs}} \mid w)$ at polynomially many parameter values w. By Condition 9.11, each such evaluation requires marginalizing over h:

$$p(x_{\mathrm{obs}} \mid w) = \sum_h \frac{\prod_C \psi_C(x_C, h_C;\, w)}{Z(w)}.$$

This marginalization is a sum-product computation on G whose complexity is governed by $\mathrm{tw}(G)$.

Step 3 (Tractability transfer). The learner $\mathcal{A}$ runs in polynomial time (by hypothesis) and must perform the evaluations of Step 2. Therefore $Z(w)$ is evaluated in polynomial time.

Note on the argument structure. This is not a pure output-extraction reduction (one cannot directly read $Z(w_0)$ at a specific w_0 from the oracle's output $p(x_{n+1} \mid x^n)$, because the predictive integrates over w). Rather, the argument combines the uniqueness theorem (which constrains *what* the oracle computes) with analysis of the Bayes predictive (which constrains *how* it must compute it) and the polynomial runtime (which forces all internal operations, including $Z(w)$ evaluations, to be tractable). □

Remark 9.16 (The backward reduction eliminates future algorithms). A forward argument ("every known method for singular learning needs Z") is vulnerable to the objection that an unknown future algorithm might avoid Z. The combined uniqueness-plus-analysis argument is stronger: it says that any oracle achieving the Watanabe-optimal rate *must* be computing the Bayes predictive (by uniqueness), and the Bayes predictive *requires* Z (by analysis of graphical models with latents). The impossibility is not "we don't know how to avoid Z" but rather "achieving the optimal rate while avoiding Z contradicts Watanabe's uniqueness theorem."

Remark 9.17 (Comparison of the four routes). Four derivation chains now lead to the same partition-function bottleneck:

Route	**Input**	**Bridge**
Exact (PKD)	exact sufficiency + ind. support	exponential family
Approximate (BvM)	non-degenerate Fisher + DQM + prior support	asymp. exponential family
Singular (Watanabe)	latent variables + Bayes comp. + optimal learning	backward reduction to Z
Fully observed	comparative Bayes revision + anchor	odds reduction to Z

All four converge at the partition function, after which Marx/Kwisthout force bounded treewidth and Theorem 7.2 gives the throughput ceiling. The four routes are complementary: the PKD route covers exact inference on regular models; the BvM route covers approximate inference on regular models; the singular route covers latent-variable models with degenerate Fisher information; the comparative route covers the fully observed residual case when belief revision exposes posterior-odds updates across hypotheses. Together they address the standard Bayesian forms of belief revision used for intelligent decision under uncertainty. What remains outside scope are systems that do not perform quality-controlled Bayesian revision at all: heuristic, discriminative, or merely empirically successful learners without posterior-odds semantics.

9.4 Fully observed singular models: the comparative-revision route

The v4 residual gap was the fully observed singular case. There the likelihood has no hidden-variable sum:

$$p(x \mid w) = \frac{\prod_C \psi_C(x_C; w)}{Z(w)}.$$

Likelihood ratios can therefore cancel the partition function, so the latent-variable argument of Theorem 9.15 no longer applies. Nevertheless, if the system performs tractable *comparative Bayesian belief revision* over competing hypotheses, then posterior odds reveal free-energy differences directly.

Condition 9.18 (Fully observed model)**.** The system is fully observed: the likelihood has no hidden-variable marginalization term,

$$p(x \mid w) = \frac{\prod_C \psi_C(x_C; w)}{Z(w)}.$$

Equivalently, uncertainty is over parameters or hypotheses, not over hidden world-state variables.

Condition 9.19 (Comparative Bayesian revision)**.** There is a polynomial-time procedure that, for arbitrary relevant hypothesis pairs w, w_0 and observations x, computes the posterior log-odds update

$$\Delta(w, w_0; x) := \log \frac{\pi(w \mid x)}{\pi(w_0 \mid x)} - \log \frac{\pi(w)}{\pi(w_0)}$$

to the accuracy required by the approximate partition-function predicate.

Condition 9.20 (Partition anchor)**.** There is a reference parameter w_0 for which the log-partition $A(w_0) = \log Z(w_0)$ is known or tractably computable. In natural families, w_0 often corresponds to the independent/no-interaction model.

Theorem 9.21 (Comparative revision implies partition-function tractability)**.** *Let a system satisfy Conditions 9.18, 9.19, and 9.20. Then the system's approximate partition function is tractable: $Z(w)$ can be evaluated to polynomial accuracy in polynomial time.*

Proof. Write

$$S_w(x) := \sum_C \log \psi_C(x_C;\, w), \qquad A(w) := \log Z(w).$$

By Condition 9.18,

$$\log p(x \mid w) = S_w(x) - A(w).$$

Hence for any pair w, w_0,

$$\log \frac{p(x \mid w)}{p(x \mid w_0)} = \big(S_w(x) - S_{w_0}(x)\big) - \big(A(w) - A(w_0)\big).$$

But Bayesian posterior odds update equals the same likelihood-ratio term:

$$\Delta(w, w_0; x) = \log \frac{p(x \mid w)}{p(x \mid w_0)}.$$

Therefore the revision oracle of Condition 9.19 reveals the free-energy difference

$$A(w) - A(w_0) = \big(S_w(x) - S_{w_0}(x)\big) - \Delta(w, w_0; x).$$

The score terms $S_w(x)$ and $S_{w_0}(x)$ are explicit from the clique potentials, so the right-hand side is tractable. Condition 9.20 gives $A(w_0)$, hence $A(w)$ itself. Exponentiating recovers $Z(w)$ to the required accuracy. □

Remark 9.22 (Why this closes the residual fully observed gap). Theorem 9.21 does not require latent variables or Watanabe-optimal asymptotics. It instead targets the exact form of belief revision left open by Remark 9.12: purely parametric uncertainty in a fully observed model. If such a system revises beliefs in a genuinely Bayesian comparative sense, then posterior odds already encode the free-energy differences needed to recover the partition function.

Remark 9.23 (Black-box vs. white-box reductions). The comparative route is cleaner than the Watanabe route. Theorem 9.21 is a true black-box reduction: one recovers $A(w) - A(w_0)$ directly from the oracle's output (posterior odds updates). By contrast, Theorem 9.15 constrains what the learner must compute by combining uniqueness with analysis of its internal Bayesian computations. The two routes are complementary: the Watanabe route handles latent-variable singular models; the comparative route handles the fully observed residual case.

9.5 Partition function tractability

The previous subsections establish, by four routes, that tractable inference forces partition-function computation: exactly via exponential family structure for systems performing exact inference (Theorem 9.1), asymptotically via exponential family structure for systems performing approximate inference (Theorem 9.8), via the backward reduction for latent-variable singular models (Theorem 9.15), and via comparative Bayesian revision for the fully observed residual case (Theorem 9.21). We now characterize when the partition function is tractable — covering both exact and approximate computation.

Definition 9.24 (Exponential family on a graph)**.** Let G be an undirected graph on N vertices. The *exponential family* on G is the collection of probability measures:

$$p_\eta(\theta) = \exp\Bigg(\sum_{c\in\text{cliques}(G)} \eta_c \cdot T_c(\theta_c) - A(\eta) \Bigg),$$

where $\{T_c\}$ are measurable sufficient statistics indexed by cliques of G, $\eta = \{\eta_c\}$ are natural parameters, and

$$A(\eta) = \log \int \exp\Bigg(\sum_{c\in\text{cliques}(G)} \eta_c \cdot T_c(\theta_c) \Bigg)\, d\mu(\theta)$$

is the *log-partition function* (normalizing constant).

Remark 9.25 (The log-partition function as the central computational object). All inference quantities — marginals, conditionals, MAP estimates, posterior expectations — are functionals of $A(\eta)$ and its derivatives. The gradient $\nabla A(\eta)$ gives expected sufficient statistics; the Hessian gives Fisher information. Tractable inference is equivalent to tractable evaluation of A and its derivatives at arbitrary points in the natural parameter space. Note that specific models on a high-treewidth graph may have tractable partition functions (e.g., Gaussian graphical models reduce to a log-determinant, computable in $O(N^3)$ regardless of treewidth). The necessity result requires hardness for *all* positive potentials on G, not just one specific model — justified by the architecture argument (Remark 9.10).

Theorem 9.26 (Bounded treewidth characterization of tractable inference)**.** *Let G be an undirected graph on N vertices.*

- *(a)* ***Sufficiency.*** *If* $\text{tw}(G) \le k$*, then the log-partition function $A(\eta)$ and all marginals of p_η on G are computable in time $O(N \cdot d^{k+1})$ for effective dimensionality d, via the junction tree algorithm [17].*
- *(b)* ***Exact necessity.*** *Assume* FPT $\neq$ #W[1]. *If $A(\eta)$ is computable in time polynomial in N for all choices of sufficient statistics $\{T_c\}$ and all natural parameters η, then* $\text{tw}(G) = O(1)$.
- *(c)* ***Approximate necessity.*** *Assume ETH (the Exponential Time Hypothesis: 3-SAT on n variables cannot be solved in time $2^{o(n)}$). If there exists a polynomial-time algorithm that, for any choice of strictly positive clique potentials $\{\psi_c\}$ on G, computes the log-partition function $A = \log \sum_\theta \prod_c \psi_c(\theta_c)$ to within additive error* 1*, then* $\text{tw}(G) = O(1)$.

Proof. Part (a) is the junction tree algorithm: given a tree decomposition of width k, the partition function factors along the junction tree, each clique integral involves at most $k+1$ coordinates, and the total cost is $O(N \cdot d^{k+1})$. This is purely continuous — no discrete structures are required [17].

Part (b). For arbitrary measurable sufficient statistics T_c, one may choose T_c to approximate indicator functions: for any finite set of configurations $\{x_1, \ldots, x_m\} \subset \Theta_c$ and any $\epsilon > 0$, there exist sufficient statistics concentrating the measure within ϵ-balls around the configurations. In the limit, the continuous partition function $\int \exp(\sum_c \eta_c T_c)\, d\mu$ converges to a discrete sum $\sum_x \exp(\sum_c \eta_c f_c(x_c))$, which is a #CSP partition function on G.

By the results of Kwisthout, Bodlaender, and van der Gaag [14], computing the partition function (equivalently, the probability of evidence in a Bayesian network) is #W[1]-hard parameterized by treewidth. Under the assumption FPT $\neq$ #W[1], this computation has no algorithm running in time $f(k) \cdot N^{O(1)}$ for unbounded k. Since the continuous family subsumes the discrete case, polynomial-time evaluation of $A(\eta)$ for all potential choices on G implies bounded treewidth.

Part (c). Given any CSP instance Φ on G with constraint functions $f_c \colon \Theta_c \to \{0,1\}$, define softened positive potentials:

$$\psi_c(\theta_c) = \begin{cases} 1 & \text{if } f_c(\theta_c) = 1 \text{ (constraint satisfied)}, \\ \varepsilon & \text{if } f_c(\theta_c) = 0 \text{ (constraint violated)}, \end{cases}$$

where $\varepsilon = e^{-4}/|\Theta|^N$ (choosing $|\Theta|$ as the domain size per variable, assuming finite domains for the reduction; the continuous case follows by approximation as in part (b)). All potentials are strictly positive, so the model satisfies our positivity requirement.

Let $Z_\varepsilon = \sum_\theta \prod_c \psi_c(\theta_c)$. If Φ is satisfiable, each satisfying assignment θ^* contributes $\prod_c \psi_c(\theta_c^*) = 1$ to Z_ε, so $Z_\varepsilon \geq 1$ and $A_\varepsilon = \log Z_\varepsilon \geq 0$. If Φ is unsatisfiable, every assignment violates at least one constraint, so each contributes at most ε, giving $Z_\varepsilon \leq |\Theta|^N \cdot \varepsilon = e^{-4}$ and $A_\varepsilon \leq -4$.

An algorithm outputting $\hat{A}$ with $|\hat{A} - A_\varepsilon| \leq 1$ yields:
- satisfiable: $\hat{A} \geq A_\varepsilon - 1 \geq -1$;
- unsatisfiable: $\hat{A} \leq A_\varepsilon + 1 \leq -3$.

These ranges are disjoint, so the algorithm decides CSP on G. By Marx [19], under ETH, CSP on constraint graphs of treewidth k cannot be solved in time $f(k) \cdot N^{o(k/\log k)}$ for any computable function f. A polynomial-time algorithm for all positive potentials on G therefore forces $\mathrm{tw}(G) = O(1)$. □

Remark 9.27 (Statement vs. proof). Theorem 9.26 is applied after the exponential-family regime has been reached: exactly via Theorem 9.1 for exact inference, asymptotically via Theorem 9.8 for approximate inference, via the backward reduction (Theorem 9.15) for singular models with latent variables, and via posterior-odds reduction (Theorem 9.21) for fully observed comparative revision. The *statement* refers to continuous objects: a graph (topology), an exponential family (information geometry), the log-partition function (normalization), and treewidth (topological invariant). The *proofs* of necessity import hardness from the discrete case: #W[1]-hardness for exact evaluation (part b), ETH-hardness for approximate evaluation (part c). Both reductions exploit the same structural fact: continuous families are strictly richer than discrete ones, so hardness in the discrete case transfers upward.

Remark 9.28 (Variational characterization). An equivalent formulation uses the Gibbs variational principle:

$$A(\eta) = \sup_\mu \left\{ \sum_c \eta_c \cdot \mathbb{E}_\mu[T_c] + H(\mu) \right\},$$

a convex optimization over probability measures [28]. For bounded treewidth, the optimization decomposes along the junction tree. For unbounded treewidth, the marginal polytope — the set of realizable expected sufficient statistics — has no compact representation, and the optimization becomes intractable. The marginal polytope's complexity is controlled by the treewidth: this is the geometric content of the necessity direction.

Remark 9.29 (The complexity assumptions are necessary). The necessity directions require complexity-theoretic assumptions: FPT $\neq$ #W[1] for exact necessity (part b) and ETH for approximate necessity (part c). This is unavoidable: the claim "tractability forces structure" is inherently complexity-theoretic — without it, one cannot rule out that an ingenious algorithm could somehow exploit unbounded treewidth. Both assumptions are standard and widely accepted:

- FPT $\neq$ #W[1]: the counting analogue of FPT $\neq$ W[1], used in Grohe's theorem for decision CSPs.
- ETH: 3-SAT requires $2^{\Omega(n)}$ time. Implies P $\neq$ NP and FPT $\neq$ W[1], and is the standard assumption for quantitative lower bounds in parameterized complexity.

The throughput bound itself (Theorem 7.2) remains unconditional — it is the *necessity of bounded treewidth* that requires a complexity assumption, not the bound's validity.

9.6 Combined necessity: class-level corollaries

The necessity corollaries are stated at the *class level*: they apply to a set $\mathcal{S}$ of continuous systems, and conclude that a uniform treewidth bound k holds across the entire class.

Definition 9.30 (Continuous system class)**.** A *continuous system class* is a set $\mathcal{S}$ of continuous systems. Since each system bundles its own N, a class may contain systems with different numbers of coordinates.

Remark 9.31 (Class-level predicates). For a class $\mathcal{S}$, we define:

- *$\mathcal{S}$ is recursively enumerable*: the induced class of effective CI graphs can be algorithmically listed.
- *$\mathcal{S}$ has bounded treewidth*: $\exists k,\ \forall\, sys \in \mathcal{S},\ \mathrm{tw}(\mathcal{G}_{\mathrm{eff}}(sys)) \leq k$.
- *$\mathcal{S}$ has all exponential family*: $\forall\, sys \in \mathcal{S},\ \mathrm{IsExponentialFamily}(sys)$.
- *$\mathcal{S}$ has all tractable partition*: $\forall\, sys \in \mathcal{S},\ \mathrm{HasTractablePartitionFunction}(sys)$.
- *$\mathcal{S}$ has all latent variables*: $\forall\, sys \in \mathcal{S},\ \mathrm{HasLatentVariables}(sys)$.
- *$\mathcal{S}$ has all tractable Watanabe-optimal learning*: $\forall\, sys \in \mathcal{S}$, there exists a polynomial-time learner achieving the Watanabe-optimal rate.
- *$\mathcal{S}$ has all fully observed models*: $\forall\, sys \in \mathcal{S},\ \mathrm{IsFullyObservedModel}(sys)$.
- *$\mathcal{S}$ has all tractable comparative revision*: $\forall\, sys \in \mathcal{S}$, there exists a polynomial-time posterior-odds revision procedure.
- *$\mathcal{S}$ has all partition anchors*: $\forall\, sys \in \mathcal{S}$, there is a tractable reference value $A(w_0)$.

And similarly for the asymptotic/approximate variants. The class-level bounded-treewidth conclusion is non-trivial because k must be uniform across all N values in the class.

Remark 9.32 (The recursive-enumerability condition). The hardness reductions of Marx (2010) and Kwisthout et al. (2010) enumerate target instances at each graph size; this requires the graph class to be recursively enumerable. The condition excludes only pathological classes definable by diagonalisation or oracle arguments. Every class that can be described algorithmically — including all classes arising from parameterized physical architectures — satisfies it.

Remark 9.33 (Why class level is necessary). For a single system with N coordinates, the conclusion $\exists k,\ \mathrm{tw}(\mathcal{G}_{\mathrm{eff}}) \leq k$ is trivially satisfied by $k = N - 1$ (put all vertices in one bag). The non-trivial content is that one bound k works uniformly across a class of systems with varying N.

Corollary 9.34 (Treewidth is constrained, not free)**.** *Under the axioms, a recursively enumerable class of systems performing tractable inference has uniformly bounded treewidth. Four derivation chains establish this:*

Exact inference *(PKD route, no asymptotics):*

$$\underbrace{\textit{Axioms 1–3}}_{\textit{unconditional}} \xrightarrow{\textit{Thm 9.1}} \textit{exp. family} \xrightarrow{\textit{Thm 9.26(b)}} \underbrace{\textit{bounded tw}}_{\textit{needs FPT}\neq\#\textit{W[1]}} \xrightarrow{\textit{Thm 7.2}} \textit{throughput } O(R_{\max}).$$

Approximate inference *(BvM route, asymptotic):*

$$\underbrace{\textit{Axioms + Conds. 9.3,9.4}}_{\textit{regular regime + prior support}} \xrightarrow{\textit{Thm 9.8}} \underbrace{\textit{asymp. exp. family}}_{\textit{as } n\to\infty} \xrightarrow{\textit{Thm 9.26(c)}} \underbrace{\textit{bounded tw}}_{\textit{needs ETH}} \xrightarrow{\textit{Thm 7.2}} \textit{throughput } O(R_{\max}$$

Singular models *(Watanabe route, backward reduction):*

$$\underbrace{\textit{Conds. 9.11,9.13 + optimal learning}}_{\textit{latent variables}} \xrightarrow{\textit{Thm 9.15}} \textit{tractable } Z \xrightarrow{\textit{Thm 9.26(c)}} \underbrace{\textit{bounded tw}}_{\textit{needs ETH}} \xrightarrow{\textit{Thm 7.2}} \textit{throughput } O($$

Fully observed singular models *(comparative route):*

$$\underbrace{\textit{Conds. 9.18,9.19,9.20}}_{\textit{comparative Bayesian revision}} \xrightarrow{\textit{Thm 9.21}} \textit{tractable } Z \xrightarrow{\textit{Thm 9.26(c)}} \underbrace{\textit{bounded tw}}_{\textit{needs ETH}} \xrightarrow{\textit{Thm 7.2}} \textit{throughput } O(R_{\max}).$$

The throughput bound (rightmost arrow) is unconditional in all four chains. The approximate chain requires non-degeneracy and local prior support (Conditions 9.3 and 9.4) and an asymptotic regime. The singular chain replaces both with a latent-variable condition (Condition 9.11), a Bayesian computation condition (Condition 9.13), and uses the backward reduction (Theorem 9.15) instead of the BvM bridge. The comparative chain closes the fully observed residual case by assuming tractable posterior-odds revision and one tractable anchor partition value.

Theorem 9.35 (Exact-inference necessity (class level))**.** *Assume* FPT $\neq$ #W[1]. *Let $\mathcal{S}$ be a recursively enumerable class of continuous systems such that every member has exact sufficiency and a tractable partition function. Then there exists k such that for all $sys \in \mathcal{S}$,*

$$\mathrm{tw}(\mathcal{G}_{\mathrm{eff}}(sys)) \le k \qquad \textit{and} \qquad \mathcal{I}(sys) \le (k+1)R_{\max}(sys).$$

Proof. By Theorem 9.1, exact sufficiency yields exponential-family structure for each system. By Theorem 9.26(b), there exists a uniform k such that every system in the class has $\mathrm{tw}(\mathcal{G}_{\mathrm{eff}}) \le k$. Apply Theorem 7.2 to each system. □

Theorem 9.36 (Approximate-inference necessity (class level))**.** *Assume ETH. Let $\mathcal{S}$ be a recursively enumerable class of continuous systems such that every member satisfies the non-degeneracy condition (Condition 9.3), has local prior support (Condition 9.4), and admits a tractable approximate partition function. Then there exists k such that for all $sys \in \mathcal{S}$,*

$$\mathrm{tw}(\mathcal{G}_{\mathrm{eff}}(sys)) \le k \qquad \textit{and} \qquad \mathcal{I}(sys) \le (k+1)R_{\max}(sys).$$

Proof. By Theorem 9.8, non-degeneracy together with local prior support yields the required asymptotic exponential-family regime. By Theorem 9.26(c), there exists a uniform k. Apply Theorem 7.2 to each system. □

Theorem 9.37 (Singular-model necessity (class level))**.** *Assume ETH. Let $\mathcal{S}$ be a recursively enumerable class of continuous systems such that every member has latent variables (Condition 9.11), satisfies the Bayesian computation condition (Condition 9.13), and admits a polynomial-time learner achieving the Watanabe-optimal generalization rate. Then there exists k such that for all $sys \in \mathcal{S}$,*

$$\mathrm{tw}(\mathcal{G}_{\mathrm{eff}}(sys)) \le k \qquad \textit{and} \qquad \mathcal{I}(sys) \le (k+1)R_{\max}(sys).$$

Proof. By Theorem 9.15, each system in $\mathcal{S}$ has a tractable approximate partition function. By Theorem 9.26(c), there exists a uniform k. Apply Theorem 7.2 to each system. □

Theorem 9.38 (Fully observed comparative-revision necessity (class level))**.** *Assume ETH. Let $\mathcal{S}$ be a recursively enumerable class of continuous systems such that every member is fully observed (Condition 9.18), satisfies tractable comparative Bayesian revision (Condition 9.19), and has a partition anchor (Condition 9.20). Then there exists k such that for all $sys \in \mathcal{S}$,*

$$\mathrm{tw}(\mathcal{G}_{\mathrm{eff}}(sys)) \le k \qquad \textit{and} \qquad \mathcal{I}(sys) \le (k+1)R_{\max}(sys).$$

Proof. By Theorem 9.21, each system in $\mathcal{S}$ has a tractable approximate partition function. By Theorem 9.26(c), there exists a uniform k. Apply Theorem 7.2 to each system. $\square$

Remark 9.39 (Per-instance instantiation). If $sys \in \mathcal{S}$, then the class-level k immediately gives a per-instance throughput bound. The Lean formalization includes explicit per-instance corollaries (`main_exact_instance`, `main_approx_instance`) obtained by instantiating from the class-level result.

Remark 9.40 (The exact route is a genuine bridge). Exact sufficiency places the system in the exponential-family regime, after which the partition-function necessity theorem forces bounded treewidth. Exact inference does not provide an escape from the bottleneck.

Remark 9.41 (Why the approximate route is load-bearing). Exact sufficiency is an idealization: real systems typically perform approximate inference. Theorem 9.36 is therefore the load-bearing necessity statement for realistic systems. The role of Bernstein–von Mises is precisely to bridge from approximate inference in regular physical regimes with local prior support to the asymptotic exponential-family setting where the bounded-treewidth necessity theorem applies.

Remark 9.42 (Approximate necessity is asymptotic). The approximate-inference route is an asymptotic necessity statement. Its force begins once the posterior is in the asymptotic regime captured by the Bernstein–von Mises bridge. During transient periods, Theorem 7.2 still constrains information flow unconditionally, but the necessity route to bounded treewidth may not yet bind.

Remark 9.43 (Architecture, not one lucky model). For the necessity argument, the key object is the architecture encoded by $\mathcal{G}_{\text{eff}}$. The right quantifier for necessity is not "there exists an easy model on this graph," but rather whether the architecture can support tractable inference across the models it may actually have to realize.

Remark 9.44 (No escape within the present theorem package). Taken together, Theorems 7.2, 9.35, 9.36, 9.37, and 9.38 state the current no-escape claim precisely. If a system already has bounded treewidth, the throughput bottleneck is immediate. If one tries to evade by appealing to tractable exact inference, Theorem 9.35 forces bounded treewidth under $\text{FPT} \neq \#\text{W}[1]$. If one appeals to tractable approximate inference in the regular regime with local prior support, Theorem 9.36 forces bounded treewidth under ETH. If one abandons regularity entirely and works with singular models, Theorem 9.37 forces bounded treewidth under ETH, provided the models have latent variables and the learner achieves the Watanabe-optimal rate. If one instead uses fully observed singular models with purely parametric uncertainty, Theorem 9.38 still forces bounded treewidth provided the system performs tractable comparative Bayesian revision. The only genuine remaining escape is therefore to abandon quality-controlled Bayesian belief revision altogether: heuristic, discriminative, or otherwise non-Bayesian systems lie outside the current theorem package.

9.7 Quantitative treewidth ceiling

The class-level corollaries above show that uniform tractability forces $k = O(1)$. A natural question is whether a single system with N state coordinates, willing to spend *more* than polynomial time, can push its treewidth higher. The relevant quantitative obstruction comes from Marx [19], who proves that under ETH, decision CSP on a graph of treewidth k requires time $N^{\Omega(k/\log k)}$. Since evaluating the log-partition function is at least as hard as decision (by the reduction in Theorem 9.26(c)), the same lower bound applies: evaluating the log-partition function on a graph of treewidth k cannot be done in time $N^{o(k/\log k)}$ uniformly over all positive potentials.

Remark 9.45 (Scaling the learned model). The class-level result together with Marx's ETH lower bound yields a two-tier picture as the system's state representation grows. Inverting the lower bound $N^{\Omega(k/\log k)}$: a time budget of N^c forces $k \le O(c \log c)$, and a quasi-polynomial budget of $N^{O(\log N)}$ forces $k \le O(\log N \cdot \log\log N)$.

Time budget	Max treewidth	Throughput ceiling
Polynomial $N^{O(1)}$	$O(1)$	$(k{+}1) \cdot R_{\max}$ (constant k)
Quasi-polynomial $N^{O(\log N)}$	$O(\log N \cdot \log\log N)$	$O(\log N \cdot \log\log N) \cdot R_{\max}$

Using the standard $O(N \cdot d^{k+1})$ junction-tree cost (where d is the effective per-coordinate state-space size), logarithmic treewidth in N is the threshold for polynomial-time scaling: if $k = O(\log N)$, inference remains polynomial; once k grows asymptotically faster than $\log N$, the cost becomes superpolynomial, and at $k = \Theta(N)$ it is exponential. Accordingly, the throughput ceiling cannot be expanded by letting structure grow without also paying a very sharp computational price.

10 Representation Adequacy and Environmental Structure

The product decomposition $\theta \in \Theta_1 \times \cdots \times \Theta_N$ is not unique: the same total state space can be factored in multiple ways, yielding different CI graphs and different bounds. This section establishes that the *environment's* structure determines a canonical factorization, resolving the apparent ambiguity.

Definition 10.1 (Environmental degrees of freedom)**.** Let $\mathcal{E} = E_1 \times \cdots \times E_d$ denote the environment's state space, factored into its independently perturbable degrees of freedom. The *environmental coupling graph* $\mathcal{G}_{\text{env}}$ has vertex set $\{1, \dots, d\}$ with an edge (j, ℓ) whenever there exist environment states under which E_j and E_ℓ are not conditionally independent given all other DOFs.

Definition 10.2 (Representation adequacy)**.** A system with state $\theta \in \Theta_1 \times \cdots \times \Theta_N$ *adequately represents* an environment with DOFs $E_1, \dots, E_d$ if:

(R1) For each DOF E_j, there exists a subset $S_j \subseteq \{1, \dots, N\}$ such that θ_{S_j} determines the system's belief distribution over E_j.

(R2) Distinct DOFs involve at least one distinct coordinate: $j \neq \ell \implies S_j \not\subseteq S_\ell$ and $S_\ell \not\subseteq S_j$.

Proposition 10.3 (Environmental lower bound on treewidth)**.** *Under representation adequacy (Definition 10.2),* $\text{tw}(\mathcal{G}_{\text{eff}}) \ge \text{tw}(\mathcal{G}_{\text{env}})$.

Proof. If environmental DOFs E_j and E_ℓ are coupled (edge in $\mathcal{G}_{\text{env}}$), then observations exist under which the system's beliefs about E_j and E_ℓ are not conditionally independent. By (R1), these beliefs are encoded in θ_{S_j} and θ_{S_ℓ}. The conditional dependence implies an edge in $\mathcal{G}_{\text{eff}}$ between some coordinate in S_j and some coordinate in S_ℓ. The map $j \mapsto S_j$ induces a graph homomorphism from $\mathcal{G}_{\text{env}}$ to a minor of $\mathcal{G}_{\text{eff}}$ (contracting each S_j to a vertex). Since treewidth is monotone under minors, $\text{tw}(\mathcal{G}_{\text{eff}}) \ge \text{tw}(\mathcal{G}_{\text{env}})$. $\square$

Remark 10.4 (The factorization is not free). A system that lumps all DOFs into a single coordinate ($N = 1$) has $\text{tw} = 0$ and bound R_1 (the single coordinate's rate), but cannot selectively update individual DOFs: every observation changes the entire state. A system that separates DOFs into distinct coordinates can update selectively, but its CI graph reflects the environment's couplings, and its treewidth is at least that of the environment's coupling graph. The product structure is determined, up to refinement, by the environment itself.

11 Effective Throughput and the Model Mismatch Tax

The throughput bound $(k{+}1) \cdot R_{\max}$ constrains *raw* throughput: the information rate $\mathcal{I}(t)$ between observation and action processes. A system's *effective* throughput — the rate at which information about the true environment state reaches its actions — may be substantially lower.

Definition 11.1 (Effective throughput rate)**.** Let $e(t)$ denote the true environment state at time t. The *effective throughput rate* is:

$$\mathcal{I}_{\text{eff}}(t) = \limsup_{\Delta t \to 0^+} \frac{1}{\Delta t} I\big(e(t);\ a_{[t,t+\Delta t]} \mid \theta(t)\big),$$

the rate at which information about the environment state reaches the actions.

Proposition 11.2 (Three-way rate bound)**.** *For the chain $e(t) \to o(t) \to \theta(t) \to a(t)$:*

$$\mathcal{I}_{\text{eff}}(t) \ \le\ \min\big\{\, \mathcal{R}(e \to o),\ \ (k{+}1) \cdot R_{\max} \,\big\},$$

where $\mathcal{R}(e \to o)$ is the information rate from environment to observation process.

Proof. The first term: by the DPI on $e(t) \to o(t) \to a(t)$, the effective throughput rate cannot exceed the rate at which the sensor reveals the environment. The second term: Theorem 7.2. □

Remark 11.3 (Interpretation)*.* Effective throughput is bounded by the *minimum* of two bottlenecks:

(i) The *observation channel*: how quickly the sensor reveals environment state.
(ii) The *inference bottleneck*: how quickly the system routes information from observation to action.

A system is sensor-limited when $\mathcal{R}(e \to o) < (k{+}1) \cdot R_{\max}$ and inference-limited when the inequality is reversed.

Definition 11.4 (Model mismatch tax)**.** The *mismatch tax rate* is $\mathcal{I}(t) - \mathcal{I}_{\text{eff}}(t)$: the throughput consumed by structural errors in the system's CI graph.

Remark 11.5 (Sources of mismatch)*.* When $\mathcal{G}_{\text{eff}} \neq \mathcal{G}_{\text{env}}$:

(i) *Missing edges* (system assumes independence that doesn't exist): the system ignores real correlations. Updates based on the wrong factorization produce posteriors that diverge from reality. Some of the raw throughput carries *wrong* information — mis-calibrated responses to observations the system interprets through an incorrect independence structure.
(ii) *Extra edges* (system assumes dependencies that don't exist): the system wastes capacity tracking phantom correlations. Separator coordinates carry redundant information, correlated with each other for no environmental reason. The effective treewidth is higher than needed, and the excess rate capacity is structural overhead.

A system whose CI graph exactly matches the environment's coupling structure ($\mathcal{G}_{\text{eff}} \cong \mathcal{G}_{\text{env}}$) pays zero mismatch tax: every bit per second through the separator carries genuine environmental information.

12 Connection to Data-Rate Control Theory

The rate formulation connects directly to classical results in networked control.

Theorem 12.1 (Nair–Evans data rate theorem [20])**.** *The minimum data rate required to stabilize a linear system $\dot{x} = Ax$ over a finite-capacity channel is $\sum_i \max(0, \mathrm{Re}(\lambda_i(A)))$ bits per second, where $\lambda_i(A)$ are the eigenvalues of A.*

Corollary 12.2 (Stabilizability bound)**.** *A system with treewidth k provides at most $(k{+}1) \cdot R_{\max}$ bits per second of observation-to-action throughput. It can stabilize only those environments whose Nair–Evans data rate requirement does not exceed $(k{+}1) \cdot R_{\max}$. Environments with sufficiently many unstable modes, or with sufficiently fast instabilities, require higher treewidth to track.*

Remark 12.3 (Quantitative complexity-demand matching). Corollary 12.2 gives a precise, quantitative version of the informal claim that "more complex environments require more complex models": the required model complexity (treewidth) is determined by the data-rate demand of the environment's unstable dynamics. The maximum environment complexity a system can handle is:

$$\sum_i \max(0, \mathrm{Re}(\lambda_i)) \;\leq\; (k{+}1) \cdot R_{\max}.$$

Each additional unit of treewidth adds $R_{\max}$ bits per second of stabilizable environment complexity.

13 Specialization to Discrete Systems

The general rate bound recovers all discrete-time, finite-domain results as a special case.

Corollary 13.1 (Discrete per-step bound)**.** *Let a system operate in discrete time $t \in \{0, 1, 2, \ldots\}$ with finite domains $\Theta_i = D_i$, $|D_i| < \infty$. Set $C_i = \log |D_i|$ and $C_{\max} = \max_i C_i$. Then:*

$$I(o_t;\, a_t \mid \theta_{t-1}) \;\leq\; (k+1) \cdot C_{\max}.$$

Proof. Apply Corollary 7.3 with $s = t-1$, $t-s = 1$. In discrete time, $R_i = C_i$ per step (each coordinate absorbs at most $\log |D_i|$ bits per step), so:

$$I(o_t;\, a_t \mid \theta_{t-1}) = I(o_{[t-1,t]};\, a_{[t-1,t]} \mid \theta(t{-}1)) \leq (k+1) \cdot C_{\max} \cdot 1.$$

□

Remark 13.2 (Reduction to the companion paper). When the clique potentials approach 0/1 indicators (softened constraint indicators), the construction degenerates to a CSP:

- The CI graph $\mathcal{G}(q)$ is the constraint graph.
- $C_i = \log |D_i|$, so the throughput bound becomes $(k{+}1) \cdot \log |D_{\max}|$ bits per step.

In the companion paper [12], the *language* of actuation strings is shown to be a $(k{+}1)$-MCFL (multiple context-free language). The information-theoretic and language-theoretic bounds coincide: the number of string fragments a grammar nonterminal juggles (at most $k{+}1$) equals the number of separator coordinates, and each fragment carries at most $\log |D_{\max}|$ bits per symbol.

14 Related Work

Bounded rationality. Herbert Simon's theory of bounded rationality [23, 24] established the foundational observation that cognitive agents operate under computational constraints and must satisfice rather than optimize. Our result gives structural content to Simon's insight: the treewidth parameter k quantifies the width of the bottleneck through which all observation-to-action information must pass. Bounded rationality is not merely a lack of optimization — it is a structural limitation on the rate at which the environment can be tracked.

Universal intelligence. Hutter's AIXI [11] defines optimal behavior as a Bayesian mixture over all computable hypotheses. With full generality the corresponding CI graph is complete ($\mathrm{tw} = N-1$), yielding a vacuously large bound. Any computable approximation of AIXI must commit to a specific CI graph with some finite treewidth k, and our bound then applies. The gap between AIXI and its approximations is precisely the gap between unrestricted treewidth and the bounded treewidth forced by tractability — quantified as $(N-k-1) \cdot R_{\max}$ bits per second of lost throughput.

Smale's eighteenth problem. Smale's eighteenth problem [25] asks: *What are the limits of intelligence, and of AI?* We contribute a rate-theoretic partial answer: under the stated conditions, the observation-to-action bandwidth of any tractable system is characterized by its position on a throughput curve parameterized by treewidth. The limitation is structural and substrate-independent.

Graphical models and inference. The role of treewidth in tractable probabilistic inference traces to the junction tree algorithm of Lauritzen and Spiegelhalter [17] and Pearl [21]. The graphical model framework of Lauritzen [16] and the variational formulation of Wainwright and Jordan [28] provide the mathematical framework connecting graph structure to inference complexity. The necessity of bounded treewidth for tractable inference was established by Kwisthout, Bodlaender, and van der Gaag [14]. Our contribution is to observe that the same treewidth that governs computational tractability also governs the information-theoretic throughput ceiling.

Data-rate control theory. The Nair–Evans data rate theorem [20] establishes that stabilizing a linear system over a finite-capacity channel requires a minimum data rate determined by the system's unstable eigenvalues. Corollary 12.2 connects this classical result to our throughput bound: a system with treewidth k can stabilize only environments whose data-rate demand does not exceed $(k+1) \cdot R_{\max}$.

CSP tractability (companion paper). The companion paper [12] establishes that for a recursively enumerable class of CSP-based embodied systems, uniform tractable belief revision with arbitrary constraint relations forces bounded treewidth (Grohe [6]), and the resulting behavior lies in a $(k+1)$-multiple context-free language (Engelfriet [5]). The discrete result and the continuous result share the same bottleneck; different mathematical languages express it.

Marr's levels. In Marr's [18] terminology, the present result operates at the *computational level*: it constrains the rate at which a system can convert observations into actions, not the algorithm it uses. The bound applies regardless of the algorithm (representation level) or physical substrate (implementation level).

15 Discussion

15.1 The CI graph as the central structural object

The entire bound is determined by $\mathcal{G}_{\text{eff}}$: its treewidth determines k, and the coordinate rates determine $R_{\max}$. When the system evolves by coupled dynamics, the sparsity pattern of the coupling directly yields $\mathcal{G}_{\text{eff}}$. In the smooth case, the Fisher information matrix (Definition 6.1) provides a Riemannian geometric realization. For exponential families, Fisher sparsity exactly reproduces the CI graph (Remark 6.2). The structural result needs only the combinatorial object: a graph derived from conditional independence under a positive model.

15.2 The effective CI graph

Every system has an effective CI graph (Remark 2.7). For exact Bayesian inference, $\mathcal{G}_{\text{eff}} = \mathcal{G}(q)$ (Theorem 2.8(a)). For graph-respecting algorithms, $\mathcal{G}_{\text{eff}} \subseteq \mathcal{G}_{\text{comp}}$. For dense algorithms, $\mathcal{G}_{\text{eff}}$ may equal K_N. The throughput bound is therefore as informative as the system's information-routing structure is sparse: small treewidth yields a sharp bottleneck, while dense routing yields a correspondingly looser bound.

15.3 No inference strategy circumvents the bound

The bottleneck is in the information routing, not the inference algorithm. The data processing inequality is the reason: if $o(t) \to \theta_C(t) \to a(t)$ is a Markov chain (which the separator property of $\mathcal{G}_{\text{eff}}$ guarantees), then the information rate from observations to actions is bounded by the rate through the separator. This holds for *any* processing of $o(t)$ — exact Bayesian update, variational approximation, MCMC sampling, gradient descent, a lookup table, or a neural network. The inequality is not "it is hard to do better"; it is "it is *impossible* to do better."

A procedure that bypasses the graph structure operates with respect to a denser effective model, and the bound applies to that model's treewidth. There is no escape: either the system respects its model's CI structure (and is rate-bounded by its treewidth), or it effectively has a denser model (and is rate-bounded by that model's treewidth).

15.4 Parallelization does not help

State update under new observations is non-monotonic (new evidence can invalidate previous beliefs). By the CALM theorem [9], non-monotonic distributed computation requires serial coordination. Parallel hardware does not bypass the serial bottleneck at the inference phase, which is where the treewidth constraint binds.

15.5 Architectural implications for real-time inference

The bound constrains which computational architectures can support tractable real-time belief revision. An architecture with effective treewidth k can perform CI-preserving inference in time $O(N \cdot \exp(k))$ [17]. For this to be tractable (polynomial in N), we need $k = O(\log N)$.

Dense architectures — transformers with global attention, fully connected networks, attention mechanisms with unrestricted context — have effective treewidth $\Theta(N)$. Every unit attends to (or depends on) every other, yielding a complete effective CI graph. Inference cost is then $\exp(\Theta(N))$: these architectures cannot support real-time belief revision at scale.

This does not mean dense architectures are useless — it means they occupy a specific role. Offline training, pattern completion, and amortized proposal generation do not require CI-preserving inference. But the real-time perception-action loop must update beliefs on every observation, subject to CI-preservation, and therefore requires an architecture whose computation graph is sparse. The treewidth of that sparse graph determines the throughput rate of the loop.

Remark 15.1 (Transformers as components within sparse architectures). A transformer with N attention heads has effective CI graph K_N and treewidth $N-1$. The throughput rate bound is $N \cdot R_{\max}$ — the total state information rate — which is correct but uninformative. The quadratic attention cost $O(N^2)$ is precisely the cost of maintaining treewidth $N-1$.

Because their forward passes operate tractably on these complete graphs, Theorem 9.36 implies that transformers cannot be fully expressive over arbitrary distributions on N variables. Their tractability is achieved precisely because the attention mechanism confines them to a structured, low-dimensional functional manifold—sacrificing the ability to faithfully represent arbitrary joint distributions for computational speed. This is the trilemma at work: dense graph, tractable inference, limited representation.

Transformers can serve as components *within* a structured architecture (e.g., as local processing modules within a sparse global graph), but the global architecture must itself be sparse for tractable real-time inference.

15.6 Universal approximation does not imply tractable expressiveness

The Universal Approximation Theorem [3, 10] guarantees that sufficiently wide neural networks can approximate any continuous function on a compact domain. This is sometimes interpreted as implying that neural networks can tractably represent arbitrary distributions, thereby escaping the treewidth bottleneck.

The interpretation conflates *existence* with *tractability*. The UAT guarantees that some parameter vector exists for which the network's output is close to the target; it does not guarantee that the network achieving this is of tractable size. Representing an arbitrary joint distribution over N variables requires width or depth that grows exponentially with N for generic targets. Once the network is constrained to polynomial size for tractable inference—as is universally the case in practice—it ceases to be a universal approximator: its representable distributions form a low-dimensional manifold in the space of all distributions over N variables. The throughput ceiling (Theorem 7.2) then applies with the effective treewidth of the network's computational graph, and the necessity result (Theorem 9.36) bounds how expressive any tractable architecture can be.

15.7 AIXI

AIXI [11] maintains a Bayesian mixture over all computable hypotheses. Its CI graph is complete ($\mathrm{tw} = N - 1$). The throughput rate bound becomes $N \cdot R_{\max}$ — vacuously large. Any implementable approximation uses a model with some finite treewidth $k \ll N$, reducing throughput rate to $(k{+}1) \cdot R_{\max}$. The gap between $N \cdot R_{\max}$ and $(k{+}1) \cdot R_{\max}$ is paid as model mismatch — information about the environment that the system's CI structure cannot route to its actuators.

15.8 What the axioms exclude

The result does not apply to:

(i) Systems with infinitely many effective degrees of freedom (violates Axiom 1; e.g., a field theory with truly infinite resolution).
(ii) Deterministic reasoners that assign zero probability density to some observations (violates Axiom 2).
(iii) Systems that do not perform probabilistic inference at all (violates Axiom 2) — though decision-theoretic arguments [27, 4] suggest any rational agent under uncertainty is effectively a probabilistic reasoner.

Remark 15.2 (Non-probabilistic systems satisfy a weaker bound). The data processing inequality $I(o(t); a(t)) \leq H(\theta_C(t))$ holds for *any* system with state θ and any separator C in any supergraph of the effective dependency graph, regardless of whether the system performs probabilistic inference. Positivity (Axiom 2) is needed only to derive the *specific* CI graph $\mathcal{G}(q)$ via the Markov property equivalence and thereby identify the *tightest* separator. Systems that lack a positive observation model still satisfy the bound, but with respect to a potentially denser effective CI graph, yielding a weaker (larger) bound. The tightest bound — from the sparsest valid CI structure — is available only to systems whose observation model is positive.

15.9 On model mismatch

The mismatch tax (§11) quantifies the throughput rate lost to structural errors in the CI graph. In practice, *every* physical environment has no true conditional independences (gravity couples everything to everything). The CI graph $\mathcal{G}_{\text{eff}}$ reflects the system's *modeling choices*, not physical reality. Every missing edge is an approximation — a coupling the system ignores for tractability — and every extra edge is wasted rate capacity. A system whose model is far from reality is doubly constrained: limited bandwidth *and* poor calibration.

15.10 The two roles of treewidth

Treewidth appears twice in this paper with distinct roles:

(i) **Throughput bound** (Theorem 7.2): Given any k, throughput $\leq (k+1) \cdot R_{\max}$. This is *unconditional*: it follows from the DPI and requires no complexity assumption, no non-degeneracy condition, and no asymptotic regime. It holds at every instant, for every system, for every inference algorithm.

(ii) **Necessity of bounded treewidth** (Corollary 9.34): Tractable inference forces $k = O(1)$ for any recursively enumerable class. For exact inference: unconditional exponential family structure (PKD) plus a complexity assumption (FPT $\neq$ #W[1]). For approximate inference: asymptotic exponential family structure (BvM, requiring Conditions 9.3 and 9.4) plus a complexity assumption (ETH), applying after the posterior has concentrated. For singular models with latent variables: the backward reduction (Theorem 9.15) plus a complexity assumption (ETH), requiring no non-degeneracy condition. For fully observed singular models: the comparative-revision reduction (Theorem 9.21) plus a complexity assumption (ETH).

The throughput bound is the paper's unconditional core. The necessity result explains *why* bounded treewidth is not merely a modeling assumption but an inevitable consequence of tractability: any architecture supporting real-time inference on the models it will encounter must commit to a graph of bounded treewidth, or face intractable computation.

15.11 The max-flow / min-cut interpretation

In the rate formulation, the bound is literally a max-flow/min-cut theorem on an information network. Information flows continuously through $\mathcal{G}_{\text{eff}}$. Each coordinate carries bounded information rate R_i. The separator is the min-cut between observation-coupled and actuation-determining coordinates. Total throughput rate $\leq$ capacity of min-cut $= (k+1) \cdot R_{\max}$. This is the natural language for the bound: it is a network capacity constraint.

15.12 Relation to the companion paper

	Discrete (companion) [12]	**General (this paper)**
State	CSP variables on finite domains	Coordinates on measurable spaces
Dependencies	Constraint hypergraph	CI graph / effective CI graph
Structure theorem	HRG characterization	Separator property
Bound type	Language class: $(k+1)$-MCFL	Throughput rate: $(k+1) \cdot R_{\max}$
Key object	Constraint graph	Effective CI graph
Same bottleneck	$\leq k+1$ string fragments	$\leq k+1$ separator coordinates
Necessity of bdd. tw	Grohe (FPT $\neq$ W[1])	Exact: PKD + partition func. (FPT $\neq$ #W[1]) Approx: BvM + partition func. (ETH) Singular: Watanabe + partition func. (ETH) Fully observed: comparative revision + partition fun (ETH)
Exp. family	N/A (direct CSP framework)	Derived from axioms (PKD/BvM)
Throughput bound	Conditional on complexity conj.	Unconditional (DPI)
Approx. route cond.	N/A (finite domains)	Conds. 9.3, 9.4
Time	Discrete steps	Continuous (discrete as corollary)

15.13 Escaping the treewidth bound: the cost of non-Bayesian inference

Every route established above—exact inference, approximate BvM, singular Watanabe, and comparative posterior odds—requires global volume integration over the parameter space. That integration forces the computation of the partition function and triggers the exponential treewidth explosion. A natural question is whether a system can escape this boundary entirely and still provide good approximation.

The answer is yes, but with quantifiable penalties. Systems performing *local point optimization*—finding a single MAP or MLE point estimate w^* via gradient descent—never compute the relative posterior mass of two isolated regions in the hypothesis space. They bypass the partition function entirely and therefore escape the #P-hardness and the exponential treewidth trap, running in polynomial time regardless of the environment's treewidth. Modern deep learning architectures trained via SGD are the canonical example.

However, such systems fall outside the scope of our four routes precisely because they abandon Bayesian volume integration, and this abandonment has well-characterized consequences in singular learning theory [29]:

1. **Sample efficiency gap.** For singular models (where the Fisher information is degenerate), Watanabe [29] showed that the Bayesian predictive distribution achieves average generalization error λ/n, where λ is the Real Log Canonical Threshold (RLCT) and n is the sample size. In the regular case, maximum likelihood achieves $d/(2n)$ where d is the parameter count; for singular models the MLE may not even be well-defined [29]. Since $\lambda < d/2$ for non-trivial singular models (and often $\lambda \ll d/2$ for deep networks), this creates a quantifiable sample-efficiency gap between Bayesian and point-estimate methods.
2. **Calibration degradation.** A point estimate collapses the posterior volume to a single configuration, discarding the uncertainty geometry. The resulting predictive distribution $q(\cdot \mid w^*)$ systematically underestimates epistemic uncertainty relative to the Bayesian predictive $\int q(\cdot \mid w)\, \pi(w \mid D)\, dw$, particularly in regions far from the training data [7, 15]. This manifests as overconfident predictions on inputs outside the training distribution.

These penalties are not incidental—they are the mathematical cost of the computational shortcut. Our theorem characterizes one side of a fundamental trade-off: Bayesian volume integration delivers optimal sample efficiency and calibrated uncertainty, but requires exponential time in the environment's treewidth. Point-optimization methods achieve tractability by abandoning volume integration, but forfeit the statistical guarantees that integration provides. The treewidth bound quantifies the computational price; singular learning theory quantifies the statistical price of avoiding it.

16 Scope of the Lean Formalization

Remark 16.1 (Included in Lean). The current continuous Lean package includes:

(i) tree-decomposition definitions and bag-width bounds,
(ii) the boundary lemmas of §4,
(iii) the throughput theorem (Theorem 7.2),
(iv) the class-level necessity corollaries for all four routes (Theorems 9.35, 9.36, 9.37, and 9.38),
(v) per-instance instantiation of the exact and approximate corollaries.

The graph lemmas and composition theorems are formalized directly. The separator theorem, the data processing inequality, the entropy inequalities, Pitman–Koopman–Darmois, Bernstein–von Mises, the Watanabe backward reduction, the comparative-revision backward reduction, and the partition-function hardness statements are imported as explicit axioms. The recursive-enumerability predicate on system classes is carried as a hypothesis throughout.

Remark 16.2 (Statement versus provenance). The clean proof states only the theorem package currently supported by the Lean development. The remarks record limiting qualifiers: architectural uniformity, asymptotic scope, and the distinction between unconditional throughput and complexity-dependent necessity.

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